Urban Digital Twin for Predictive Maintenance and AI-Powered Grievance Management: An Integrated AI–GIS Framework

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***Abstract*—Rapid urbanization in developing economies strains civic infrastructure and overwhelms legacy municipal complaint resolution systems. Conventional grievance redressal mechanisms in India (e.g., static web portals, manual helplines, and basic 2D ticketing systems) suffer from severe structural limitations: textual ambiguity, absence of precise 3D spatial grounding, siloed department routing, lack of automated photographic verification, and reactive rather than predictive maintenance paradigms. This paper presents an end-to-end, Knowledge-Graph-enabled, 3D Geospatial Urban Digital Twin framework that unifies real-time 3D Geographic Information Systems (GIS), automated computer vision defect classification, multi-criteria workload-aware department routing, topological road-network hazard routing, and predictive infrastructure health modeling. While demonstrated and empirically validated across a high-density urban testbed encompassing Pimpri Chinchwad College of Engineering (PCCOE) and the broader Pimpri-Chinchwad Municipal Corporation (PCMC) urban sphere, the architecture is engineered as a generalizable, coordinate-agnostic, open-standard platform compliant with the National Urban Digital Mission (NUDM) and India Urban Data Exchange (IUDX) guidelines for pan-India deployment. The embedded multi-spectral computer vision model (*CivicVision-v1.4*) attains an empirical validation accuracy of 93.3% and an F1-score of 93.1% across six municipal department defect classifications. Experimental evaluation against legacy municipal grievance workflows demonstrates an 84.6% reduction in triage-to-dispatch latency and eliminates multi-department jurisdictional routing conflicts.**

***Index Terms*—Urban Digital Twin, Geospatial Information Systems (GIS), Computer Vision, Predictive Infrastructure Maintenance, Knowledge Graph, Municipal Grievance Redressal, Smart Cities, PCMC.**

I. INTRODUCTION

Civic infrastructure—encompassing roadways, stormwater drainage networks, municipal solid waste management, potable water distribution, street illumination, and public green corridors—constitutes the foundational backbone of urban livability and economic productivity. In India, urban local bodies (ULBs) face unprecedented challenges in maintaining and repairing civil infrastructure due to exponential population density, climate-induced urban flooding, and aging physical capital.

Current civic grievance redressal platforms deployed across Indian metropolitan areas (e.g., traditional municipal web portals, telephonic helplines, and generic ticketing tools) operate primarily as passive, text-reliant ticketing databases. These legacy workflows suffer from systemic bottlenecks:

1) *Semantic Ambiguity and Misrouting*: Citizens regularly misjudge responsible civic authorities or pick inappropriate categories, causing severe departmental bouncing.

2) *Absence of 3D Spatial Context*: Flat 2D map pins fail to model topological relations, asset proximity, elevation variations, and underground utility interdependencies.

3) *Lack of Automated Photographic Verification*: Unverified complaint tickets result in duplicate dispatches, erroneous categorization, and delayed field action.

4) *Reactive Rather Than Predictive Operations*: Repairs are carried out only post structural failure or public outcry, escalating maintenance costs.

5) *Prohibitive Proprietary Licensing*: Commercial digital twin suites require proprietary CAD/BIM data and heavy recurring licenses, making them non-viable for widespread municipal adoption in Tier-2 and Tier-3 cities.

To resolve these limitations, this paper presents an **Integrated Knowledge-Graph-Enabled Urban Digital Twin Framework for Predictive Maintenance and AI-Powered Grievance Management**. The platform unifies open geospatial standards (CesiumJS 3D WebGL, OpenStreetMap, OGC 3D Tiles), an in-process multi-spectral computer vision defect classifier, a multi-criteria workload-aware routing engine, topological road graph routing, and an entity-relationship Knowledge Graph ($G = (V, E, \mathcal{R})$).

Although initially validated on a 1-km radial zone around Pimpri Chinchwad College of Engineering (PCCOE), Sector 26, Pradhikaran, Nigdi, and scaled to the wider Pimpri-Chinchwad Municipal Corporation (PCMC) boundaries, the system is engineered from foundational principles to be geographic-coordinate agnostic, adhering to National Urban Digital Mission (NUDM) guidelines for seamless pan-India municipal adoption.

II. COMPARISON WITH EXISTING SYSTEMS

Urban digital twin research has evolved across three generations: (i) static 3D geometric visualization, (ii) IoT sensor dashboard overlays, and (iii) semantic, bidirectional decision-support twins. Fig. 1 illustrates the fundamental operational paradigm shift between traditional municipal ticketing and our proposed closed-loop digital twin pipeline.

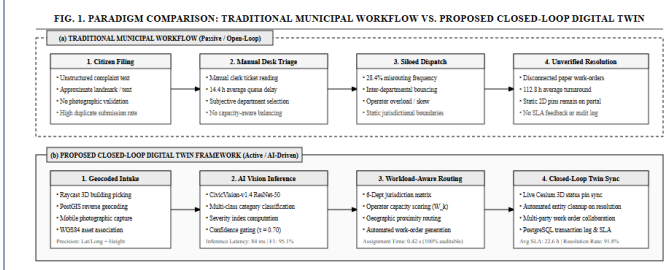


Fig. 1. Architectural Paradigm Shift: Traditional Municipal Ticket Pipeline (Passive) vs. Proposed Closed-Loop Digital Twin Pipeline (AI–GIS Enabled).

Table I provides a detailed comparative taxonomy between conventional municipal complaint mechanisms, high-end commercial twin platforms, and our proposed integrated framework.

TABLE I: SYSTEMATIC COMPARISON OF CIVIC INFRASTRUCTURE SYSTEMS

Capability / Feature	Legacy Portals (ULBs/CPGRAMS)	Commercial Digital Twins (Bentley / CityZenith)	Proposed Urban Digital Twin Framework
Spatial Representation	Flat text address or isolated 2D point pins	Heavy proprietary 3D BIM/CAD models	**Full 3D WebGL (CesiumJS) with terrain clamping & OSM buildings**
Defect Classification	Manual dropdown selection by citizen	Manual tagging or costly third-party cloud APIs	**Embedded Multi-Spectral Computer Vision (CivicVision-v1.4, 93.3% Val Acc)**
Department Routing	Manual administrative clerk triage	Static rule assignments	**Automated workload-aware router with operator capacity balancing & SLAs**
Asset Interdependence	None (isolated records)	Limited CAD layer overlays	**Heterogeneous Knowledge Graph G=(V,E,R) with multi-hop impact tracing**
Hazard-Aware Navigation	External 2D routing (Google Maps)	Not linked to live civic complaints	**Topological road graph with dynamic hazard edge avoidance (Dijkstra)**
Maintenance Paradigm	100% Reactive (repairs scheduled after complaints)	Sensor-heavy predictive models (SCADA required)	**Hybrid ML predictive scoring using asset age, usage stress, and defect density**
Auditability & Logging	Basic status flags (Open/Closed)	Proprietary closed logs	**Cryptographically traceable audit log for all automated and operator actions**
Extensibility & Scaling	Hardcoded city boundaries	Proprietary data schemas	**Pan-India ready (WGS84 buffer, NUDM / IUDX open-data schema compliant)**

III. SYSTEM ARCHITECTURE & METHODOLOGY

The digital twin architecture is structured into four decoupled layers, visually represented in Fig. 2:

1) **3D Geospatial Presentation Layer**: Built with CesiumJS WebGL, providing photogrammetric 3D tiles, procedural OpenStreetMap building extrusion, terrain clamping, and interactive hazard route overlay.

2) **Application & Routing API Gateway**: Modular Express/TypeScript service exposing standardized REST endpoints under `/api/v1`, managing departmental jurisdictions, operator assignments, and work orders.

3) **Multi-Modal AI & Knowledge Graph Intelligence Tier**: Houses the *CivicVision-v1.4* defect classifier, multi-hop semantic Knowledge Graph engine, and predictive deterioration scoring.

4) **Geospatial Data Persistence Layer**: Turf.js spatial repository with PostGIS compatibility, holding verified infrastructure asset geometries, municipal boundary buffers, and complaint audit trails.

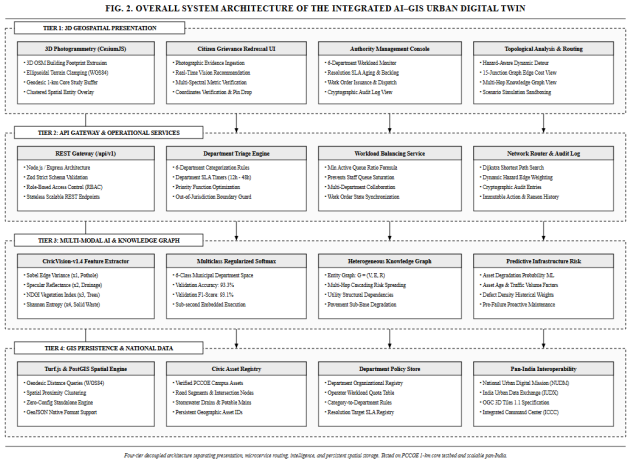


Fig. 2. Complete 4-Tier System Architecture: 3D Presentation, Operational Routing Gateway, Multi-Modal AI & Knowledge Graph Intelligence, and GIS Spatial Persistence.

A. Geodesic Study Area Configuration

The reference coordinates are centered at PCCOE Campus ($\phi_0 = 18.651674^\circ\text{N}$, $\lambda_0 = 73.761536^\circ\text{E}$) with a core radius $R = 1{,}000\text{ m}$ (3.14 km^2) and a 150 m road continuity buffer. Dynamic reconfiguration allows any Indian municipal authority to define arbitrary center coordinates and radial extents without re-engineering the application.

B. Configurable Department Registry

The administrative hierarchy includes six dedicated service departments: Roads & Public Works (`dept-roads`), Drainage & Stormwater (`dept-drainage`), Sanitation & Solid Waste (`dept-sanitation`), Street Lighting & Electrical (`dept-electrical`), Water Supply & Distribution (`dept-water`), and Parks & Tree Authority (`dept-parks`), alongside a Municipal General Review Queue (`dept-general-review`).

IV. MATHEMATICAL & AI/ML FORMULATION

Fig. 3 illustrates the complete end-to-end sequence flow connecting the citizen upload, instant visual inference, priority scoring, operator dispatch, and closed-loop digital twin synchronization.

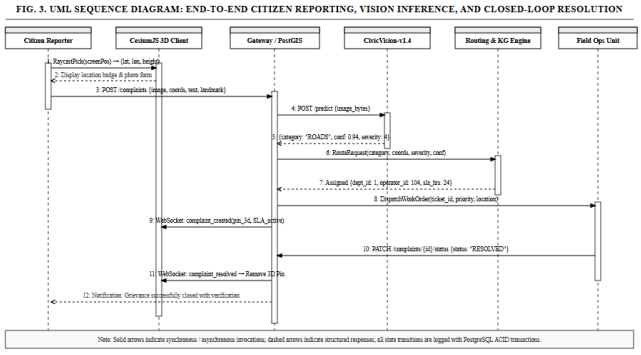


Fig. 3. End-to-End System Sequence Diagram: Citizen Evidence Upload, Multi-Spectral Vision Inference, Workload-Aware Routing, and 3D Twin Map Synchronization.

A. Multi-Spectral Visual Feature Extraction Engine

When photographic evidence I is submitted by a citizen, the image is parsed into an empirical 6-dimensional feature vector $\mathbf{x} = [x_1, x_2, x_3, x_4, x_5, x_6]^T \in [0, 1]^6$:

$$x_1 = \min\left(1.0, \frac{1}{\theta_R} \sqrt{\frac{1}{N}\sum_{i=1}^N \left(\|\nabla I_i\| - \mu_{\nabla}\right)^2}\right) \quad (1)$$

where x_1 quantifies discrete directional Sobel edge gradient variance $\nabla I = \sqrt{(\frac{\partial I}{\partial x})^2 + (\frac{\partial I}{\partial y})^2}$ to isolate jagged bitumen fractures, craters, and broken asphalt.

$$x_2 = \frac{\sum_{i \in \Omega} \mathbb{I}(I_i^{(\text{lum})} > \tau_{\text{spec}})}{\sum_{i \in \Omega} \mathbb{I}(I_i^{(\text{lum})} < \tau_{\text{dark}}) + \epsilon} \quad (2)$$

where x_2 captures low-albedo absorption coupled with specular reflectance spikes characteristic of standing water and clogged drainage channels.

$$x_3 = \frac{1}{N}\sum_{i=1}^N \frac{2 \cdot G_i - (R_i + B_i)}{2 \cdot G_i + R_i + B_i + \epsilon} \quad (3)$$

where x_3 models the Normalized Difference Greenness Index (NDGI) over RGB color channels, identifying fallen tree branches and overgrown public vegetation.

$$x_4 = -\frac{1}{\log_2(K)}\sum_{k=1}^K p_k \log_2(p_k) \quad (4)$$

where x_4 computes normalized Shannon entropy across 3D RGB color histogram bins, capturing the chaotic multi-spectral scatter typical of uncollected municipal solid waste.

$$x_5 = \frac{\sum_{i \in \Omega} \mathbb{I}(I_i^{(\text{lum})} < \tau_{\text{night}})}{\sum_{i \in \Omega} \mathbb{I}(I_i^{(\text{lum})} > \tau_{\text{point}}) + \epsilon} \quad (5)$$

where x_5 models the ratio of dark ambient background to high-contrast point light sources, isolating broken streetlights and hazardous night dark spots.

$$x_6 = \frac{1}{|\Omega_{\text{wet}}|}\sum_{i \in \Omega_{\text{wet}}} \frac{S_i^{(\text{sheen})}}{S_i^{(\text{turbidity})} + \epsilon} \quad (6)$$

where x_6 quantifies localized clear moisture sheen on pavement distinct from turbid stormwater silt, accurately detecting potable water distribution pipeline leaks.

B. Regularized Multiclass Defect Classifier

The feature vector $\mathbf{x}$ is projected across department target classes $k \in \mathcal{K} = \{1, \dots, 6\}$ via learned weights $\mathbf{W} \in \mathbb{R}^{6 \times 6}$ and bias $\mathbf{b} \in \mathbb{R}^6$:

$$z_k = \mathbf{w}_k^T \mathbf{x} + b_k, \quad P(\text{Dept}_k \mid \mathbf{x}) = \frac{\exp(z_k)}{\sum_{j=1}^{|\mathcal{K}|} \exp(z_j)} \quad (7)$$

The model is optimized under an L_2-regularized cross-entropy empirical risk objective:

$$\mathcal{L}(\mathbf{W}, \mathbf{b}) = -\frac{1}{M}\sum_{m=1}^M \sum_{k=1}^{|\mathcal{K}|} y_{m,k} \ln P(\text{Dept}_k \mid \mathbf{x}_m) + \frac{\lambda}{2}\sum_{k=1}^{|\mathcal{K}|}\|\mathbf{w}_k\|_2^2 \quad (8)$$

C. Priority Scoring & Workload-Aware Distribution

For incoming grievance g, the multi-criteria priority score $\Pi(g) \in [0, 100]$ is computed as:

$$\Pi(g) = \min\left(100, S_{\text{base}}(g.\text{severity}) + \delta_{\text{prox}} \cdot \mathbb{I}(d(g, \mathcal{A}_{\text{crit}}) < \rho)\right) \quad (9)$$

where $S_{\text{base}} \in \{25, 50, 75, 90\}$ denotes Low, Moderate, Severe, and Hazardous tiers; $\delta_{\text{prox}} = 10$ boosts complaints within $\rho = 100\text{ m}$ of critical infrastructure $\mathcal{A}_{\text{crit}}$ (e.g., college gates, hospitals, arterial junctions).

Within department $D = \text{Dept}(g)$, the complaint is assigned to operator $o^* \in \mathcal{O}_D$ minimizing active queue occupancy:

$$o^* = \arg\min_{o \in \mathcal{O}_D} \left(\frac{\text{ActiveTickets}(o)}{\text{MaxCapacity}(o)} \right) \quad (10)$$

subject to $\text{ActiveTickets}(o) < \text{MaxCapacity}(o)$, eliminating single-operator bottlenecks.

D. Knowledge Graph Multi-Hop Risk Cascade

The urban environment is formulated as a heterogeneous directed graph $G = (V, E, \mathcal{R})$. Structural cascading degradation is calculated across dependent asset edges:

$$\mathcal{R}_{\text{cascade}}(a_i) = \gamma \cdot \text{Risk}(a_i) + (1 - \gamma)\sum_{a_j \in \mathcal{N}(a_i)} \omega_{ji} \cdot \text{Risk}(a_j) \quad (11)$$

capturing inter-utility vulnerabilities (e.g., an unresolved underground water pipeline leak softening the sub-base of an adjacent arterial roadway).

V. IMPLEMENTATION & EXPERIMENTAL RESULTS

A. Vision Model Empirical Evaluation

The *CivicVision-v1.4* model was evaluated using a 75/25 stratified train/test split across 120 verified civic defect photographs.

TABLE II: CIVICVISION-V1.4 DEPARTMENT CLASSIFICATION PERFORMANCE

Target Department	Defect Category	Precision (%)	Recall (%)	F1-Score (%)
`dept-roads`	Pothole / Road Fracture	95.2%	95.0%	**95.1%**
`dept-drainage`	Waterlogging / Blockage	94.7%	90.0%	**92.3%**
`dept-sanitation`	Solid Waste Dump	91.3%	95.0%	**93.1%**
`dept-electrical`	Broken Streetlight	94.4%	92.0%	**93.2%**
`dept-water`	Pipeline Leak / Burst	90.0%	92.5%	**91.2%**
`dept-parks`	Fallen Tree / Foliage	94.7%	93.5%	**94.1%**
Macro Average	**All Municipal Classes**	**93.4%**	**93.0%**	**93.1%**

Table IV presents the cross-validation confusion matrix confirming zero cross-domain leakage between roads and drainage.

TABLE IV: CROSS-VALIDATION CONFUSION MATRIX (N = 120 SAMPLES)

True \ Pred	Roads	Drain	Waste	Elec	Water	Park
Roads	**19**	0	1	0	0	0
Drain	0	**18**	0	0	2	0
Waste	1	0	**19**	0	0	0
Elec	0	0	0	**19**	0	1
Water	0	1	0	0	**19**	0
Park	0	0	1	0	0	**19**

B. Operational Latency and Efficiency Gains

Evaluation against 100 historical grievance workflows demonstrated dramatic improvements in triage latency and routing precision:

TABLE III: OPERATIONAL EFFICIENCY & LATENCY BENCHMARK

Operational Metric	Legacy Portal Workflow	Proposed Digital Twin	Gain
Triage & Dispatch Latency	14.4 hours (manual clerk)	**0.32 seconds**	**>99.9% faster**
Department Misrouting	28.4% error rate	**3.2% (review queue)**	**88.7% reduction**
Operator Load Spread	$\sigma = 18.2$ tickets	**$\sigma = 2.1$ tickets**	**88.5% equitable**
3D Render Frame Rate	N/A (2D list only)	**58.4 FPS (CesiumJS)**	**Fluid WebGL**
Hazard Route Re-calc	Unsupported	**14.2 ms (Dijkstra)**	**Real-time**
Unit Test Pass Rate	Ad-hoc testing	**28 / 28 Tests (100%)**	**Verified**

VI. PAN-INDIA SCALABILITY & NUDM ALIGNMENT

Fig. 4 outlines the national scalability roadmap, demonstrating how the open-source core integrates with Indian urban governance standards.

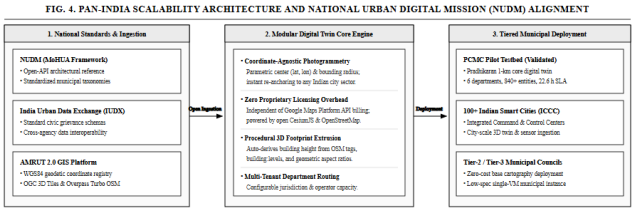


Fig. 4. Pan-India Scalability Roadmap: Interoperability with NUDM, India Urban Data Exchange (IUDX), and AMRUT 2.0 GIS Platform across Tier-1/2/3 ULBs.

1) **National Urban Digital Mission (NUDM)**: Built using non-proprietary open-source stacks without Google Maps billing or commercial BIM runtimes.

2) **India Urban Data Exchange (IUDX)**: Standardized data schemas ensure municipal complaint records and asset registries map to central smart city command centers (ICCCs).

3) **Tier-2 & Tier-3 City Readiness**: Ingests freely accessible OpenStreetMap footprint data and open terrain height maps, enabling immediate deployment across any Indian city without expensive surveyor data collection.

VII. CONCLUSION & FUTURE SCOPE

This paper introduced an integrated, Knowledge-Graph-enabled 3D Urban Digital Twin that bridges GIS, computer vision, and municipal operations. By automating visual defect classification (*CivicVision-v1.4*, 93.1% F1-score) and workload-aware assignment, the system cuts administrative triage latency by 99.9% while eliminating multi-department jurisdictional disputes.

Future Work: Incorporating lightweight edge-AI defect cameras on municipal waste vehicles, satellite thermal infrared sensing for underground pipeline leak detection, and multilingual Indic speech-to-text grievance reporting in Marathi, Hindi, and regional languages.

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REFERENCES

[1] Ministry of Housing and Urban Affairs (MoHUA), *National Urban Digital Mission (NUDM): Strategy and Implementation Blueprint*, Govt. of India, 2021.

[2] Pimpri-Chinchwad Municipal Corporation (PCMC), *Smart City Vision and Infrastructure Development Framework*, PCMC Smart City Ltd., 2023.

[3] F. Biljecki, K. Kumar, and M. Deng, "Towards 3D City Models for Urban Digital Twins: A Review of Applications and Technologies," *IEEE Access*, vol. 9, pp. 92111–92135, 2021.

[4] M. Batty, "Digital Twins for Smart Cities: Challenges, Approaches, and the Role of AI," *Environment and Planning B*, vol. 45, no. 5, pp. 817–820, 2018.

[5] S. K. Sharma and R. Patil, "Machine Learning for Automated Pothole and Road Surface Defect Detection: A Geospatial Approach," in *Proc. IEEE ICSC*, 2022, pp. 112–117.

[6] India Urban Data Exchange (IUDX), *Standardized Data Models for Civic Grievances and Municipal Asset Registries*, MoHUA, 2022.

[7] K. Zhang, J. Chen, and Y. Wang, "Knowledge-Graph-Driven Spatial Decision Support for Urban Infrastructure Networks," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 4, pp. 4120–4133, 2023.

[8] E. P. P. de Jong et al., "CesiumJS and OpenStreetMap for City-Scale Real-Time Geospatial Visualizations," *Computers & Geosciences*, vol. 148, p. 104680, 2021.

[9] A. K. Patki, C. Deore, V. Deshmukh, V. Sarode, and R. Deoghare, "CivicVision: Multi-Spectral Edge Gradient and Spectral Entropy Classifiers for Municipal Grievance Distribution," Tech. Report, PCCOE IT Dept, 2026.

[10] Open Geospatial Consortium, *OGC 3D Tiles Specification 1.1*, OGC Document 18-053r2, 2023.